Merchant Needs, Usage & Product-Concept Validation

Research design • merchant segmentation • hypotheses • test methodology • safeguards • management handoff

19 October 2026 • Version 1.0 for pre-joining research • Confidential working draft

CONTROL	CURRENT STATUS
Document reference	PSS-A360-V04-01; research protocol under Volume II
Author / accountability	Pre-joining strategy research; proposed future RPB&D owner, not an official PSS appointment
Authorised research	None. No merchants approached, no interview executed, no PSS system access.
Study decision supported	Merchant segment prioritisation and concept-selection evidence for V03/S02 and F01; not pilot or launch approval.
Dependencies	V02 market baseline; V03 bank commercial validation; V05 product verification; Legal/Compliance/CISO consent/data authority.
Evidence cut-off	19 October 2026; public evidence and prior programme drafts only

Executive decision statement

The merchant research must find out not simply whether businesses can take digital payments, but whether they can recognise incoming funds, reconcile sales with balances, resolve missing/uncertain transactions, and benefit from separate business payment records. The proposed buyer of Operate + Connect is ordinarily a bank, while a merchant is the service end-user; merchant enthusiasm alone is not proof of bank willingness to pay.

Preferred research sequence: explore existing behaviours without showing any concept; observe a recalled transaction scenario; compare four neutral concept cards; then rank real pain and plausible benefit. Translate merchant-level findings into specific bank-service requirements, not arbitrary product features.

Status: protocol authored only. No representative survey, field results, usability success rate, user approval or formal institutional endorsement is claimed.

1. Research baseline and important new 2026 evidence

The Merchant Payments Observatory (DFS Lab) reports Ethiopia Wave 1 findings from 2,724 merchants, weighted to represent approximately 500,000 enterprises within its sampled target population: 58.8% acceptance, 26% rural adoption, more than 99% of digital flows into personal accounts and connectivity cited as a leading barrier (R02). These are published study findings, not PSS performance figures. NBE still describes the 2026–2030 digital payments strategy as a draft on its public portal (R01).

Critical denominator note: The Observatory's target merchant population is not the same as the draft NBE broad ~2 million MSME estimate; 58.8% must not be multiplied by NBE's estimate to produce a national merchant count. The World Bank Global Findex is a consumer/adult survey (R03), not an observational merchant interview panel.

EVIDENCE	DESIGN IMPLICATION	PROHIBITED LEAP
Existing merchant acceptance can be widespread (R02)	Ask how merchants accept, not only whether they accept.	Assume every merchant is unserved.
Personal accounts dominate reported flows (R02)	Map payment identity, records, account separation, receipts and merchant ID.	Assume merchants want or can lawfully be forced to migrate.

EVIDENCE	DESIGN IMPLICATION	PROHIBITED LEAP
Rural adoption 26% in source target (R02)	Test low-connectivity methods and urban/rural contrasts.	Treat rural as one homogeneous need.
Cash continues to compete strongly (R05–R06)	Explore perceived safety, speed, familiarity, fees and day-end handling.	Assume digital is always preferred.
Phone-based contactless has scheme/PCI conditions (R07)	Use paper/non-production storyboard only until approval.	Ask real merchants to enter sensitive card data for research.

2. Objectives and research questions

RQ01 What payment channels are used for which transaction types; what are the last three recalled payment experiences?

RQ02 How are digital payments distinguished from personal transfers, confirmed to customers, and reconciled daily?

RQ03 Which failures (delay, wrong payer, reference mismatch, chargeback, refund, intermittent network) cost merchants time or money?

RQ04 What alternative workflow, bank app, wallet, paper ledger or aggregator already solves the problem?

RQ05 What merchant-visible improvement is valuable without additional purchase of a new national platform?

RQ06 Which device, network, literacy, language and accessibility conditions affect adoption?

RQ07 Does formality, rurality, gender or merchant segment meaningfully change observed needs?

RQ08 Which concept should be carried forward to bank validation: operations, connectivity-assisted acceptance, PSS Pay-led, or current practice?

3. Falsifiable merchant hypotheses

ID	TESTABLE PROPOSITION	FALSIFICATION EXAMPLE
H01	Merchants already accepting digital payments have unresolved reconciliation pain	Counter: existing ledger or SMS process closes sales reliably with little effort.
H02	Merchant/personal account distinction causes operational difficulties	Counter: merchants report no missed or misattributed payments and no demand for separation.
H03	A transaction confirmation/exception view improves merchant comprehension	Counter: no measurable difference from existing app or paper process.
H04	Connectivity and electricity are more material than software feature gaps in rural segments	Counter: strong reliable service and preference for existing tools.
H05	A bank-sponsored support path is preferred over dealing with each wallet/provider	Counter: providers already handle complaints with acceptable speed.
H06	A unified multi-channel sales view is understood and valued	Counter: the merchant uses one channel and finds aggregation unnecessary.
H07	PSS Pay-like phone card acceptance fits a definable device-ready segment	Counter: low NFC/Android compatibility or customers cannot pay by card.
H08	Daily closing/report export reduces time or error	Counter: observed task takes no less time or increases complexity.
H09	Merchant cohorts differ enough to require distinct propositions	Counter: little meaningful need contrast after channel/use controls.
H10	Merchants will accept bank-mediated onboarding and merchant account arrangements	Counter: requirements and trust concerns make acceptance low.

4. Sample design and recruitment controls

Recommended qualitative discovery sample: 36 merchants (expand to 48 for saturation where justified); this is a planning target, not a statistically representative national survey. Recruit across urban and secondary-city/peri-urban sites and an appropriately authorised rural comparison area. Do not claim nationally estimated adoption rates from this sample.

RECRUITMENT QUOTA / CONTRAST	PLANNED N	REASON
Cash-only / predominantly cash	10	Explore genuine non-adoption and cash workflow
Wallet/QR via personal account	10	Investigate informal transfer/merchant record challenges
Formal merchant QR / bank account	8	Test transaction servicing and reporting with business-account users
POS / card / mixed-channel merchants	8	Explore device and card-payment readiness

Within every group, balance merchant types (retail groceries, market traders, services, hospitality, transportation where safe), ownership gender, days trading, revenue category (self-reported), device ownership and bank/wallet provider. Use a recruitment log to monitor diversity without recruiting only through one bank. Aim for 4–6 contextual walkthroughs and 8–12 non-production concept evaluations across contrasting groups as subsets of the 36 (not extra counted respondents).

Sampling limitations Purposive/maximum-variation sampling supports pattern discovery and hypothesis refinement, not population incidence estimates; exact rural/urban quotas, access/security and language coverage require locally approved fieldwork plan.

5. Four concept cards: neutral comparisons, rotating order

CARD	WHAT RESPONDENT SEES (HYPOTHETICAL ONLY)	COUNTERFACTUAL CHALLENGE
A — Operations & exceptions	A simple timeline shows payment received, pending or needing checking; daily totals and a bank help route.	Would an ordinary account statement or existing wallet already suffice?
B — Multi-channel sales view	Payments from permitted channels shown with payer-safe reference and downloadable daily totals.	Do you actually receive enough cross-channel payments to need this?
C — Phone/card acceptance	A merchant-capable eligible phone could accept appropriate contactless cards via authorised product.	Are the phone, customer card base, support and security prerequisites present?
D — Current practice/no new tool	Merchant continues present bank/wallet/cash workflow with improvements to training and customer support.	What is lost if there is no new application?

For each card, first ask respondent to explain what the concept does in their own words. Then ask likelihood of use, realistic frequency, expected trade-off, confusion, effort and which existing tool they would stop using. Rotate presentation order by respondent ID; do not lead with branding or promise that PSS currently offers the feature. Never solicit banking credentials or attempt live payments.

6. Behavioural research protocol

STAGE / TIME	REQUIRED ACTION	EVIDENCE TO CAPTURE
Opening • 5 min	Read approved consent in preferred language; explain independent hypothetical research and right to stop.	Anonymous participant ID, consent and choice of observation/photo separately.
Unaided reality • 12 min	Walk through last three purchases and a problematic transaction, without requesting customer identities.	Channels used, steps, existing workarounds, failure reasons and time cost.
Day-close walkthrough • 8 min	Ask how merchant checks daily takings, handles unclear settlements and customer disputes.	Process map, pain severity, frequency, bank/provider dependency.
Concept comparison • 15 min	Rotate A–D cards; read neutral scripts; non-production paper/sandbox task only.	Comprehension, completion, comments, ranked value, objections.
Commercial perceptions • 5 min	Ask who should provide support and who should pay; do not frame merchant fees as already legal.	Credible service payer hypothesis; cost sensitivity; bank ownership.
Close • 5 min	Invite corrections, reconfirm retention/withdrawal channels per approved policy.	Study notes, unresolved queries and recruitment segment.

7. Key definitions and measurements

MEASURE	DEFINITION / DENOMINATOR
Recent acceptance	Self-report: accepted at least one business payment digitally within the last 30 days. Exclude personal P2P unless for a business sale; record distinctions.
Formal merchant payment	Respondent identifies merchant QR/MID or dedicated business-merchant service; do not infer from provider logo.
Problem incidence	Count respondents reporting a specified problem in last 30 days / respondents asked; describe as sample count, not population rate.
Daily-close time	Minutes reported or observed to reconcile preceding day; annotate measurement method and incomplete records.
Task success	Participants independently reaching the correct result using approved non-production concept; denominator = participants assigned task.
Confidence	1–5 self-reported confidence in payment status after an observed task; not a validated psychometric measure.
Preferred concept	Rank-1 among A/B/C/D after exposure to rotated order; report segment counts and reasons; not purchase conversion.
Device readiness	Observed/self-reported phone OS, NFC, battery/connectivity; do not store device identifiers, SIM numbers or biometrics.
Merchant fee tolerance	Hypothetical context only; do not translate consumer preference to bank contract revenue.

8. Analysis and decision standards

Triangulation Compare merchant statements with observed or demonstrated behaviours, available public research, and bank-side V03 needs. Conflicting evidence is recorded as conflicting; never harmonise away.

Threshold policy No invented numeric acceptance target. Before approved fieldwork, pre-register practical decision rules and bank minimum-economics requirements with Research, Finance and Risk; scores will be described as qualitative until validated.

Proposed concept decision Carry forward only a concept that addresses a repeated observed problem, improves a defined task or observable service outcome over existing practice, is supported by an authorised bank buyer, and can meet PSS operator/legal/certification requirements. Any failed mandatory condition blocks proceeding.

Uncertain or no-demand outcome Recommend improve existing services/training, re-scope target segment, or discontinue a new merchant-facing module. Negative evidence is a programme success if it avoids waste.

9. Research ethics, security and authorisation

- Pre-joining period: desk research and draft interviews only. Do not contact or represent PSS to banks, merchants or vendors on PSS's behalf.
- Before recruitment: obtain written sponsor/institutional fieldwork approval, Legal and CISO-approved consent and data retention rules, and reviewed Amharic/Afaan Oromo/other language versions where applicable.
- Do not collect names, phone numbers, bank account numbers, card PAN, CVV, PIN, OTP, customer identities or real transaction screenshots. Pseudonymous survey IDs only.
- Do not record or photograph without separate explicit consent; do not store exact shop location, customer identities or payment logs unless approved as necessary with lawful basis.
- No live payments, card taps or security tests; use generic examples or approved isolated demo. Immediate stop if a real customer's sensitive data enters observation.
- Use secure PSS-approved collection/storage after appointment, tightly scoped access, incident escalation and approved retention/deletion schedule; no retention period assumed in this draft.
- Participation voluntary and not contingent on banking services. Compensation, if any, must be policy-approved and must not bias responses.

10. Fieldwork quality and disconfirmation plan

QUALITY CONTROL	PLANNED CHECK
Interviewer calibration	Pilot 2–3 interviews outside analytic sample after approval; refine terminology without changing hypothesis after data collection without logging change.
Evidence traceability	Every quote linked to pseudonymous ID, permission status, date, instrument version and researcher.
Researcher neutrality	Randomise/rotate concept order and neutral script; audit 10% of notes for leading interpretation where authorised.
Saturation	After 24 interviews, check whether new problem themes emerge in each segment; continue/expand to 48 if meaningful gaps remain.
Disconfirming cases	Actively sample businesses reporting that existing arrangements work well; document why.
Causal restraint	Do not infer that a single observed use test proves impact on commercial revenue or fraud.

11. Deliverables, roles and decision handoffs

DELIVERABLE	PRODUCT / HANDOFF	REVIEW / BLOCKER
V04-A Segment evidence report	Merchant jobs-to-be-done, behaviours, barriers and segment contradictions	Research lead; representative-claim prohibition
V04-B Process maps	Cash, transfer, merchant QR, POS, uncertain outcome and day-close workflows	Product/Operations; no credentials in maps
V04-C Concept report	Balanced A/B/C/D comparison, usability notes, rejected ideas	Research + Risk; concept design not approval
V04-D Merchant service requirements	What banks must support or configure in service catalogue	V03 bank sponsor + P01 product team

DELIVERABLE	PRODUCT / HANDOFF	REVIEW / BLOCKER
V04-E Control report	Recruitment, consent, de-identification, sampling limitations	Legal/CISO before publishing sensitive excerpts
Cross-volume amendment	V02 market fact-table update for MPO 2026 and segment methodology	Research owner + document controller

Decision handoff: V04 informs V03 bank value propositions, P01 product requirements, S02 option scores and F04/F05 economics. It is never a standalone justification to approve an investment or promise field findings.

12. Pre-appointment jobs vs after-joining jobs

CAN PREPARE NOW	REQUIRES APPOINTMENT + APPROVAL
Evidence review and cross-reference new MPO Ethiopia Wave 1 research	Official merchant outreach and recruitment
Prepare instruments, concept cards, scoring definitions and sample plan	Pilot interviews, observation, device checks, ethics-consent execution
Design pseudonymised blank evidence and analysis workbook	Handling respondent data under approved PSS controls
Prepare V02 market amendment note and V03 bank pain-point test alignment	Presenting validated evidence as PSS work; bank-facing product representations

Annex A — Sources and provenance

REF	PRIMARY / SPECIALIST MATERIAL	ACCESS AND SCOPE
R01	NBE — Draft National Digital Payments Strategy, 2026–2030	https://nbe.gov.et/ndps/ Draft; current web label as of 19 Oct 2026
R02	DFS Lab / Merchant Payments Observatory — Ethiopia Wave 1 research publications, April 2026	https://mpo.dfslab.net/research.html Merchant survey 2,724; weighted to ~500,000 enterprises; see scope cautions
R03	World Bank — Global Findex 2025, Ethiopia 2024 microdata	https://microdata.worldbank.org/catalog/7901/data-dictionary/F1 Adult demand-side evidence, not merchant population sampling
R04	GSMA — What Will It Take to Scale Merchant Payments? January 2026	https://www.gsma.com/solutions-and-impact/connectivity-for-good/mobile-for-development/gsma_resources/what-will-it-take-to-scale-merchant-payments-harnessing-the-digital-opportunity/ Cross-market context, not an Ethiopia merchant census
R05	CGAP — Merchant Payments: Adding Value for Merchants, 2019	https://www.cgap.org/research/publication/merchant-payments-adding-value-for-merchants Product research framework, not current Ethiopia statistics
R06	CGAP — Choosing a Profit Strategy for Merchant Payments, 2019	https://www.cgap.org/research/publication/choosing-profit-strategy-for-merchant-payments Pricing and value proposition principles
R07	PCI SSC — Mobile Payments on COTS (MPoC)	https://www.pcisecuritystandards.org/standards/mobile-payments-on-cots-mpoc/ Certification context; not evidence of PSS Pay approval